

Problem C. Team Division

Time Limit 2000 ms

Problem Statement

Divide N players, player 1, player 2, $\dots$, player N , into two (distinguishable) teams A and B satisfying all of the following conditions.

- Each team consists of at least one player.
- Each player belongs to either team A or team B , but not both.
- The number of players in the team that player i belongs to is at least L_i and at most R_i .

Find the number of ways to divide the players satisfying the conditions, and output the count modulo 998244353.

Two divisions are considered different if there exists a player who belongs to different teams in the two divisions.

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $1 \leq L_i \leq R_i \leq N - 1$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N
L1 R1
L2 R2
⋮
LN RN
```

Output

Output the number of valid divisions, modulo 998244353.

Sample 1

Input	Output
3 1 1 1 2 2 2	2

The following two divisions satisfy the conditions.

- Player 1: Team A , Player 2: Team B , Player 3: Team B
- Player 1: Team B , Player 2: Team A , Player 3: Team A

Since 2 modulo 998244353 is 2, output 2.

Sample 2

Input	Output
6 1 5 1 5 2 5 1 3 3 5 2 5	30

There are 30 valid divisions.